Artificial Intelligence | Computer Vision | Embedded Systems

PATCHARADANAI SOMBATSATIEN

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MSc graduate in Computer Science and Technology with an electrical engineering background and experience in embedded control software, hardware and software development, and electrical testing. Focuses on AI and computer vision, with training in deep learning and parameter-efficient adaptation. Co-author of research on astronomical instrumentation and aerosol detection.

Education Background

MSc in Computer Science and Technology | Xi'an Jiaotong University (XJTU), China 2024 - 2026 | GPA 3.63

BE in Electrical Engineering | Chiang Mai University (CMU) 2017 - 2021 | GPA 3.78

Professional Experience

Embedded Software Engineer Feb 2023 - Aug 2024

National Astronomical Research Institute of Thailand (NARIT)

- Designed and implemented embedded control software for high-speed systems used in astronomical research, working with real-time operating systems.
- Processed atmospheric LiDAR data with range correction, worked on laser power control, and visualized and corrected spectral data from a UV-to-visible spectrograph.
- Co-authored four SPIE conference papers on adaptive optics, coronagraph prototypes and opto-mechanical design, and a conference contribution on atmospheric aerosol detection using Mie LiDAR.

Electrical Test Engineer May 2021 - Jan 2023

Hana Microelectronics Group

- Troubleshot motherboards for high-frequency Keysight instruments, applying electrical testing and hardware fault diagnosis to investigate board-level issues.
- Managed electrical testing on production test lines for 50 GHz surfboard instrument motherboards (TOFINO, TOFINO-X, and WALMEA), supporting manufacturing test operations.
- Contributed to four production improvement projects covering instrument tracking, automated work-in-progress reporting, test-station notifications and semi-automatic RF spurious testing.

Technical Skills

Programming skills in Python, C++ and VHDL, with experience in embedded systems, real-time operating systems, digital logic design and FPGA-based image processing. Background in electrical testing, circuit analysis, hardware and software debugging, and programmable logic controllers. AI and data skills include machine learning, deep learning, computer vision, parameter-efficient adaptation and data handling.

Languages

Thai: Native | English: IELTS 6.5/9.0 & TOEIC 760/990 | Chinese: HSK 3, 270/300

Engineering Outreach

NECTEC Electronics Camp Instructor Oct 2018 and Dec 2019

National Electronics and Computer Technology Center

- Taught digital logic design and fundamental electronics to high-school participants in digital electronics workshops.

Mae Chan Drainage Community Service Project May 2019

Purple Lab, Chiang Mai University, with Blue Lab, University of Michigan

- Applied engineering principles to improve the Mae Chan drainage system in the community service project.
- Designed hydraulic cleaning equipment, rotating and inflatable flow-control devices, and a web-based drainage network mapping system for maintenance and water management.

Selected Research and Engineering Projects

LoRA Fusion for Open Vocabulary Segmentation

Master's dissertation | Xi'an Jiaotong University | Sep 2024 – July 2026

- Proposed a parameter-efficient, training-free domain adaptation framework for open-vocabulary segmentation, addressing the transfer of segmentation models across visual domains.
- Used similarity-guided LoRA adapter selection and dynamic fusion to compose adapters from heterogeneous pre-training. The framework aims to retain zero-shot generalization while supporting cross-domain stability and computational efficiency.
- Compared RGB images, zero-shot predictions and the proposed framework qualitatively on street scenes.

Atmospheric LiDAR Data Processing and Laser Control

NARIT | Dec 2023 - Aug 2024

- Processed Mie-scattering LiDAR measurements using range correction and worked on laser power control for atmospheric observations.
- Compared range-corrected oscilloscope measurements with Micro Pulse LiDAR profiles to examine signal behavior across altitude, including observations collected on 3 April 2024.
- The project included an Atmospheric LiDAR dashboard showing measurement profiles, a time selector, data downloads and weather information to support inspection of observation data.

UV to Visible Spectrograph Data Processing

NARIT | Jan 2024 - May 2024

- Visualized spectral data cubes and corrected image distortion to support observation of solar features with a UV-to-visible spectrograph covering 350-700 nm.
- Produced wavelength-specific views, including a solar image near 656.59 nm, and before-and-after comparisons illustrating correction of distorted spectral images.

Production Test Automation and Monitoring

Hana Microelectronics and Keysight Technologies | May 2021 - Jan 2023

- Contributed to four improvement projects supporting RF instrument testing and production operations.
- Where is my instrument: the workflow used Python and PyVISA to communicate with instruments, connect to a database and send email notifications for instrument tracking.
- Orion Automatic WIP: automated a workflow connecting the Orion server and Gondor to Excel output for work-in-progress reporting.
- Weekly Test Station Update: delivered LINE Notify updates covering preventive maintenance, equipment status and test-station readiness.
- Semi-automatic Spurious Test: contributed to conversion of a manual RF spurious test to a semi-automatic workflow using Python and PyVISA to control a signal analyzer.

Vehicle Counting at Road Intersections Using Video Data

Undergraduate senior project | Chiang Mai University | 2020

- Developed an image-processing system to count vehicles traveling along individual routes at a road intersection using video data.
- Collected route-level vehicle counts as primary traffic data for investigating congestion patterns and supporting future traffic-management studies.

Research Projects

Evanescent Wave Coronagraph and Wavefront Correction

NARIT | Co-author of three SPIE conference papers | 2024

- Co-authored research on optical design for the Thai National Telescope, preliminary closed-loop adaptive optics performance, and non-common path aberration correction in the Evanescent Wave Coronagraph testbed.
- The aberration-correction study used a deformable mirror and the science camera as a wavefront sensor. Mirror commands were optimized by maximizing image variance at the detector to correct residual bench aberrations and compare laboratory performance with theoretical expectations.

Flexible Mount Optimization for Space Telescope Metallic Mirrors

NARIT and collaborating institutions | Co-author | 2024

- Co-authored a finite element analysis study of flexible mounting pads for lightweight metallic mirrors, addressing optical stability and mechanical performance from manufacturing to orbital operation.
- Developed automation scripts to streamline repetitive CAD modeling tasks in SolidWorks, reducing manual effort, improving workflow efficiency, and ensuring consistency across design iterations.

Publications and Conference Contributions

1. C. Pothipor, S. Awiphan, T. Supasri, T. Thomrungpiyathan, **P. Sombatsatien**, W. Kumtan, W. Kaima, *"Detection of atmospheric aerosol layer using Mie LiDAR system"*, JT-AND 2025
2. A. Ratkata, A. Berdeu, **P. Sombatsatien**, K. Bannajak, M. Alagao, Michel Tallon, E. Thiébaut, M. Langlois, T. Thomrungpiyathan, T. Lepine, S. Awiphan, S. Poshyachinda, W. Rujopakarn, C. Buisset, *"Preliminary closed-loop performance of the adaptive optics for the Evanescent Wave Coronagraph prototype"*, SPIE 2024 DOI: 10.1117/12.3019455
3. P. Pongam, W. Wanajaroen, M. Alagao, T. Lepine, T. Thomrungpiyathan, **P. Sombatsatien**, A. Ratkata, S. Awiphan, *"Completed optical design and performance analysis of the evanescent wave coronagraph prototype for the Thai National Telescope"*, SPIE 2024 DOI: 10.1117/12.3019607
4. N. Mahawan, P. Panpang, N. Muangnin, C. Bourgenot, C. Atkins, R. Snell, M. Phlernjai, W. Wanajaroen, **P. Sombatsatien**, P. Chartsiriwattana, *"Optimization of flexible opto-mechanical mounts design in metallic mirror for space-based telescope using finite element analysis"*, SPIE 2024 DOI: 10.1117/12.3018676
5. M. Alagao, A. Berdeu, M. Tallon, E. Thiébaut, M. Langlois, P. Pongam, T. Lepine, S. Awiphan, M. Ridsdill-Smith, A. Ratkata, **P. Sombatsatien**, T. Thomrungpiyathan, S. Poshyachinda, W. Rujopakarn, C. Buisset, *"First results of the non-common path aberration correction in the Evanescent Wave Coronagraph testbed"*, SPIE 2024 DOI: 10.1117/12.3019936

Honors and Awards

- **2024:** Full Scholarship for Master's Degree, Xi'an Jiaotong University. IT Foundation under the Initiative of HRH Princess Maha Chakri Sirindhorn; covered tuition and living expenses in Xi'an, China.
- **2021:** First Place, True 5G Robotic Bootcamp, True Corporation. Represented Chiang Mai University and developed an intelligent robotic solution.
- **2019:** First Place, Digital Design Camp x CMU, Chiang Mai University and Design Gateway. Re-implemented an FPGA image processing application and presented the solution.
- **2019:** Outstanding Engineering Student Scholarship, Chiang Mai University. Recognized for academic performance in the Faculty of Engineering.
- **2014:** Silver Award, National Mathematics Competition, Education Service Area Office.
- **2011:** Gold Award, Mathematics Academic Competition, Chiang Mai Primary Education Service Office

Training and Professional Development

- **Mar 2025:** From NLP to LLM Building Smart Systems, Xi'an Jiaotong University. Hands-on training in NLP, large language models and intelligent system design.
- **Oct 2024:** Big Data and Machine Learning Workshop, Xi'an Jiaotong University. Practical training in big data analytics and machine learning.
- **Mar 2021:** Shopee Code League, Shopee. Programming competition and data science training covering algorithm design and business challenges.
- **Dec 2020:** True 5G Robotic Bootcamp, True Corporation. Training in robotic application development and business model design; developed and presented a team project.